Gesture Recognition Stacking Blocks

1. Gesture Recognition Instructions

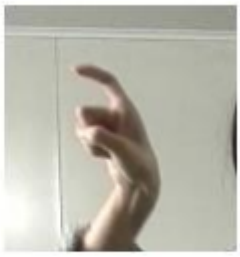
1. This routine uses gesture recognition based on Baidu Smart Cloud platform service, which provides 50,000 free uses per day, for educational purposes only, not for commercial use. If long-term use is required, please purchase related services. Our company is not responsible for any issues.
2. For KEY application instructions and API documentation, please refer to the previous section of the course content.
3. Supported gestures and example images for gesture recognition

序号	手势名称	classname	示例图
1	数字 1（原食指）	One	
2	数字 5（原掌心向前）	Five	
3	拳头	Fist	

4	OK	OK	
5	祈祷	Prayer	
6	作揖	Congratulation	
7	作别	Honour	
8	单手比心	Heart_single	
9	点赞	Thumb_up	

10	Diss	Thumb_down	
11	我爱你	ILY	
12	掌心向上	Palm_up	
13	双手比心 1	Heart_1	
14	双手比心 2	Heart_2	
15	双手比心 3	Heart_3	

16	数字 2	two	
17	数字 3	three	
18	数字 4	four	
19	数字 6	six	
20	数字 7	seven	
21	数字 8	eight	

22	数字 9	nine	
23	Rock	Rock	
24	竖中指	Insult	

2. Experiment Setup

Place the experiment setup according to the following: yellow blocks in the yellow area, red blocks in the red area, green blocks in the green area, blue blocks in the blue area.

3. Code Content

- Start Docker

After entering the main board desktop, open the terminal and run the following command to start the Dofbot corresponding container:

```
./Docker_Ros.sh
```

Then enter the following command in the docker terminal to start jupyter-lab,

```
cd
jupyter-lab --allow-root
```

Then enter the main board's IP address plus :9999 in the browser,

```
IP:9999         // Example: 192.168.1.11:9999
```

Code path: /root/Dofbot/6.AI_Visual/2.gesture_stack.ipynb

```
#bgr8 to jpeg format conversion
import enum
import cv2
def bgr8_to_jpeg(value, quality=75):
    return bytes(cv2.imencode('.jpg', value)[1])
```

```
# Import related modules

import   threading
import   time
from   Arm_Lib import Arm_Device
#   Create robotic arm object
Arm   = Arm_Device()
time.sleep(.1)
```

```
#   Define gesture recognition function part
import   cv2
import   time
import   demjson
import   pygame

from   aip import AipBodyAnalysis
from   aip import AipSpeech
from   PIL import Image, ImageDraw, ImageFont
import   numpy
import   ipywidgets.widgets as widgets

#   For specific gestures, please refer to the official documentation
https://ai.baidu.com/ai-doc/BODY/4k3cpywrv

hand={'One':'数字1','Two':'数字2','Three':'数字3','Four':'数字4',

      'Five':'数字5', 'Six':'数字6','Seven':'数字7',
```

```
      'Eight':'数字8','Nine':'数字9','Fist':'拳头','Ok':'OK',

      'Prayer':'祈祷','Congratulation':'作揖','Honour':'作别',

      'Heart_single':'比心心','Thumb_up':'点赞','Thumb_down':'Diss',

      'ILY':'我爱你','Palm_up':'掌心向上','Heart_1':'双手比心1',

      'Heart_2':'双手比心2','Heart_3':'双手比心3','Rock':'Rock',

      'Insult':'竖中指','Face':'脸'}

#   Replace the key below with your own

"""   Body Analysis   APPID AK SK """

APP_ID   = '18550528'

API_KEY   = 'K6PWqtiUTKYK1fYaz13O8E3i'

SECRET_KEY   = 'IDBUII1j6srF1XVNDX32I2WpuwBWczzK'

client   = AipBodyAnalysis(APP_ID, API_KEY, SECRET_KEY)

g_camera   = cv2.VideoCapture(0)
g_camera.set(3,   640)
g_camera.set(4,   480)
g_camera.set(5,   30)  #Set frame rate
g_camera.set(cv2.CAP_PROP_FOURCC,   cv2.VideoWriter.fourcc('M', 'J', 'P', 'G'))
g_camera.set(cv2.CAP_PROP_BRIGHTNESS,   40) #Set brightness -64 - 64  0.0
g_camera.set(cv2.CAP_PROP_CONTRAST,   50) #Set contrast -64 - 64  2.0
g_camera.set(cv2.CAP_PROP_EXPOSURE,   156) #Set exposure value 1.0 - 5000  156.0

ret,   frame = g_camera.read()
```

```
#   Define camera display component

image_widget   = widgets.Image(format='jpeg', width=600, height=500)  #Set
camera display component

display(image_widget)

image_widget.value   = bgr8_to_jpeg(frame)
```

```
#   Define function to display Chinese text

def   cv2ImgAddText(img, text, left, top, textColor=(0, 255, 0), textSize=20):

    if (isinstance(img, numpy.ndarray)):  # Check if it's OpenCV image type

        img =   Image.fromarray(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))

    # Create an object that can draw on the given image

    draw = ImageDraw.Draw(img)
```

```
    # Font format

    fontStyle = ImageFont.truetype(

        "simhei.ttf", textSize,   encoding="utf-8")

    # Draw text

    draw.text((left, top), text, textColor,   font=fontStyle)

    # Convert back to OpenCV format

    return cv2.cvtColor(numpy.asarray(img),   cv2.COLOR_RGB2BGR)
```

```
#   Define variable parameters for different positions
look_at   = [90, 164, 18, 0, 90, 90]
p_top   = [90, 80, 50, 50, 270]

p_Yellow   = [65, 22, 64, 56, 270]
p_Red   = [118, 19, 66, 56, 270]

p_Green   = [136, 66, 20, 29, 270]
p_Blue   = [44, 66, 20, 28, 270]

p_layer_4   = [90, 76, 40, 17, 270]
p_layer_3   = [90, 65, 44, 17, 270]
p_layer_2   = [90, 65, 25, 36, 270]
p_layer_1   = [90, 48, 35, 30, 270]

p_push_over_1   = [90, 90, 5, 0, 90, 150]
p_push_over_2   = [90, 90, 0, 50, 90, 150]

#Define block grabbing status
yellow_grabbed   = 0
red_grabbed   = 0
green_grabbed   = 0
blue_grabbed   = 0

#Define gesture recognition count

Count_One   = 0
Count_Two   = 0
Count_Three   = 0
Count_Four   = 0
Count_Fist   = 0
```

```
#   Define robotic arm movement function, control servos 1-6 simultaneously, p=
[S1,S2,S3,S4,S5,S6]

def   arm_move_6(p, s_time = 500):

    for i in range(6):

        id = i + 1

        Arm.Arm_serial_servo_write(id, p[i],   s_time)
```

```
        time.sleep(.01)

    time.sleep(s_time/1000)



#   Define robotic arm movement function, control servos 1-5 simultaneously, p=
[S1,S2,S3,S4,S5]

def   arm_move(p, s_time = 500):
    for i in range(5):
        id = i + 1

        if id == 5:

            time.sleep(.1)

            Arm.Arm_serial_servo_write(id,   p[i], int(s_time*1.2))

        elif id == 1 :

            Arm.Arm_serial_servo_write(id,   p[i], int(3*s_time/4))
        else:
            Arm.Arm_serial_servo_write(id,   p[i], int(s_time))
        time.sleep(.01)
    time.sleep(s_time/1000)



#   Define block clamping function, enable=1: clamp, =0: release

def   arm_clamp_block(enable):
    if enable == 0:
        Arm.Arm_serial_servo_write(6, 60,   400)

    else:
        Arm.Arm_serial_servo_write(6, 130,   400)

    time.sleep(.5)
```

```
#Number function definition

def   number_action(index):

    if index == 1:

        # Grab the yellow block

        arm_move(p_top, 1000)

        arm_move(p_Yellow, 1000)

        arm_clamp_block(1)

#         time.sleep(.5)
```

```
        arm_move(p_top, 1000)
    elif index == 2:
        # Grab the red block
        arm_move(p_top, 1000)
        arm_move(p_Red, 1000)
        arm_clamp_block(1)
        arm_move(p_top, 1000)
    elif index == 3:
        # Grab the green block
        arm_move(p_top, 1000)
        arm_move(p_Green, 1000)
        arm_clamp_block(1)
        arm_move(p_top, 1000)
    elif index == 4:
        # Grab the blue block
        arm_move(p_top, 1000)
        arm_move(p_Blue, 1000)
        arm_clamp_block(1)
        arm_move(p_top, 1000)



def   put_down_block(layer):


    if layer == 1:
        arm_move(p_layer_1, 1000)
        arm_clamp_block(0)
        arm_move_6(look_at, 1000)
    elif layer == 2:
        arm_move(p_layer_2, 1000)
```

```
        arm_clamp_block(0)
        arm_move_6(look_at, 1000)
    elif layer == 3:
        arm_move(p_layer_3, 1000)
        arm_clamp_block(0)
        arm_move_6(look_at, 1000)
    elif layer == 4:
        arm_move(p_layer_4, 1000)
        time.sleep(.1)
        arm_clamp_block(0)
        arm_move_6(look_at, 1000)
#   Push over blocks
def   push_over_block():
    arm_move_6(p_push_over_1, 1000)
    time.sleep(.2)
    arm_move_6(p_push_over_2, 1000)
    time.sleep(.1)
    arm_move_6(look_at, 1000)
    time.sleep(1)
    global g_layer
    g_layer = 0
```

```
#Move the robotic arm to the camera forward-looking position
arm_move_6(look_at,   1000)
time.sleep(1)
```

```
global   g_state_arm
g_state_arm   = 0
global   g_layer
g_layer   = 0
```

```python
def   ctrl_arm_move(index):

    global g_layer

    g_layer = g_layer + 1

    if g_layer >= 5:

        g_layer = 1

    arm_clamp_block(0)

    if index == 1:

        number_action(index)

        put_down_block(g_layer)

    elif index == 2:

        number_action(index)

        put_down_block(g_layer)

    elif index == 3:

        number_action(index)

        put_down_block(g_layer)

    elif index == 4:

        number_action(index)

        put_down_block(g_layer)

    elif index == 5:

        time.sleep(1)

        push_over_block()


    global g_state_arm

    g_state_arm = 0
```

```python
def   start_move_arm(index):
    # Start robotic arm control thread
    global g_state_arm

    if g_state_arm == 0:

        closeTid = threading.Thread(target =   ctrl_arm_move, args = [index])

        closeTid.setDaemon(True)

        closeTid.start()

        g_state_arm = 1
```

```python
try:

    Arm.Arm_Buzzer_On(1)

    s_time = 300

    Arm.Arm_serial_servo_write(4, 10, s_time)

    time.sleep(s_time/1000)

    Arm.Arm_serial_servo_write(4, 0, s_time)

    time.sleep(s_time/1000)

    Arm.Arm_serial_servo_write(4, 10, s_time)

    time.sleep(s_time/1000)

    Arm.Arm_serial_servo_write(4, 0, s_time)

    time.sleep(s_time/1000)



    while True:

        """1. Take photo   """

        ret, frame = g_camera.read()



        """ 2. Call gesture recognition   """

        raw =   str(client.gesture(image_widget.value))

        text = demjson.decode(raw)

        try:

            res =   text['result'][0]['classname']
```

```
        except:
    #           print('Recognition result: Nothing recognized~' )
    #           img = cv2ImgAddText(frame, "Not recognized", 250, 30, (0, 0 ,
255), 30)
            img = frame
        else:
    #           print('Recognition result:' + hand[res])
    #           img = cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 , 0), 30)
            if res == 'One':
                Count_One = Count_One + 1
                Count_Two = 0
                Count_Three = 0
                Count_Four = 0
                Count_Fist = 0
                if Count_One >= 3:
                    print('识别结果: ' + hand[res])
                    img = cv2ImgAddText(frame,   hand[res], 250, 30, (0, 255 ,
0), 30)
                    if yellow_grabbed == 0:
                        start_move_arm(1)
    #                     global yellow_grabbed
                        yellow_grabbed = 1
            elif res == 'Two':
                Count_Two = Count_Two + 1
                Count_Three = 0
                Count_Four = 0
                Count_Fist = 0
                if Count_Two >= 3:
                    print('识别结果: ' + hand[res])
                    img =   cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 ,
0), 30)
```

```
                    if red_grabbed == 0:
                        start_move_arm(2)
    #                     global red_grabbed
                        red_grabbed = 1
            elif res == 'Three':
                Count_Three = Count_Three + 1
                Count_Two = 0
                Count_Four = 0
                Count_Fist = 0
                if Count_Three >= 3:
                    print('识别结果: ' + hand[res])
                    img = cv2ImgAddText(frame,   hand[res], 250, 30, (0, 255 ,
0), 30)
                    if green_grabbed == 0:
                        start_move_arm(3)
    #                     global green_grabbed
                        green_grabbed = 1
              elif res == 'Four':
                Count_Four = Count_Four + 1
                Count_Two = 0
                Count_Three = 0
                Count_Fist = 0
                if Count_Four >= 3:
                    print('识别结果: ' + hand[res])
                    img = cv2ImgAddText(frame,   hand[res], 250, 30, (0, 255 ,
0), 30)
                    if blue_grabbed == 0:
                        start_move_arm(4)
    #                     global blue_grabbed
                        blue_grabbed = 1
```

```
            elif res == 'Fist': #Fist

                Count_Fist = Count_Fist + 1

                Count_One = 0

                Count_Two = 0

                Count_Three = 0

                Count_Four = 0

                if Count_Fist >= 3:

                    print('识别结果: ' + hand[res])

                    img =   cv2ImgAddText(frame, hand[res], 250, 30, (0, 255 ,
0), 30)

                    start_move_arm(5)

                    yellow_grabbed = 0

                    red_grabbed = 0

                    green_grabbed = 0

                    blue_grabbed = 0

                    Count_Fist = 0
            else:
                img = frame

        image_widget.value =   bgr8_to_jpeg(img)

except   KeyboardInterrupt:

    print(" Program closed! ")

    pass
```

```
If the set gesture is recognized, the robotic arm will perform the corresponding
action. When the robotic arm recognizes a number for the first time, it grabs a
block and places it on the first layer; when it recognizes a gesture number for
the second time, it grabs a block and places it on the second layer; when it
recognizes a gesture number for the third time, it grabs a block and places it on
the third layer; when it recognizes a gesture number for the fourth time, it
grabs a block and places it on the fourth layer. The color of the block grabbed
by the robotic arm each time is determined by the recognized number: number 1
represents yellow, number 2 represents red, number 3 represents green, number 4
represents blue. When this number is recognized again, it will not grab again
unless a fist is recognized, which will push over all blocks, clear the records,
and start over.

The gestures and corresponding actions in this routine are as follows:
```

数字1	夹取黄色区域的积木块
数字2	夹取红色区域的积木块
数字3	夹取绿色区域的积木块
数字4	夹取蓝色区域的积木块
拳头	推倒堆叠好的积木块，清空记录